



China's high-quality economic development: a study of regional variations and spatial evolution

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Abstract

Based on the current reality of unbalanced and inadequate regional economic development in China, this study constructs a comprehensive index system to assess China's economic development quality across six dimensions: innovation dynamic, coordinated development, market governance, green transformation, open economy, and people's welfare. It employs the vertical and horizontal scatter degree approach that better captures the inherent characteristics of data to dynamically measure China's high-quality economic development index (HQEDI) and unveils its spatial interaction patterns through a modified dynamic Spatial Durbin Model. Findings indicate that China's HQEDI demonstrates a persistent upward trend and an expanding cycle, with significant spatial agglomeration effects. The spatial correlation structure portrays an imbalanced distribution of HQEDI, with Shanghai and Beijing as the center, the eastern coast as the secondary center, the central and western regions occupying subordinate positions. Under various spatial weights, China's HQEDI exhibits conditional β convergence and positive spillover effect. Convergence rates decrease in the order of "Central-West-Northeast-East," with corresponding convergence cycles increasing in this order.

Keywords High-quality economic development · Dynamic evaluation · Spatial evolution · Regional differences

JEL Classification C51 · O11 · R12

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1 Introduction

In the 40 years since China's reform and opening up, China's GDP has grown by 33.54 times, with an average annual real growth rate of 9.5%, far exceeding the world's average of 2.9% during the same period (World Bank data). However, the long-term rapid economic growth has also brought about a series of inevitable problems such as imbalance in economic structure, resource depletion, environmental pollution, and lack of social welfare, which have become major obstacles to sustained and stable economic growth. These phenomena not only hinder the progress of China's economic development, but also present a common challenge for many developing countries in the realm of economic growth. The root of this issue lies in the primary economic theories that underpin the economic growth models of developing countries, specifically neoclassical economic growth theory and new economic growth theory, prioritize quantitative growth while largely neglecting improvements in the quality of the development process. Therefore, the concept of "inclusive growth" is being incorporated into the economic strategies and frameworks of countries worldwide, aiming to explore the realization of coordinated and sustainable economic development. Against this background, China has put forward a strategic concept to transition its economic development model from high-speed growth to high-quality development. High-quality economic development implies that the efficiency of economic growth primarily stems from innovation rather than just scale expansion, with few major structural imbalances, minimal negative externalities in terms of resources and the environment, and a fairer distribution of the benefits of economic development among the entire population. Its connotation adheres to the fundamental idea of inclusive growth, rooted in the challenges and practical pathways of China's economic and social development transformation. It encompasses universality, aligned with the common principles of international economic development, while also catering to the specific economic development environment and needs of China. Therefore, recognizing the actuality and structural distinctions within the high-quality development of various regional economies in China holds crucial theoretical and practical importance in comprehending the fundamental principles and policy trajectories of China's economic development strategy transformation. Moreover, it can offer insights and serve as a reference for developing countries to attain inclusive growth.

The existing research on assessing the quality of China's economic high development is primarily limited by three aspects: First, the criteria for selecting indicators to measure the quality of economic development exhibit significant arbitrariness. Second, the determination of indicator weights largely relies on subjective assessments, without taking into consideration the inherent characteristics of the data. Third, while the disparities in the quality of economic development among regions have attracted widespread attention, there is a lack of exploration on the correlation and interaction among the quality of economic development in different regions. The issue of regional economic development imbalance has been a focal point in the China's academic community for a long time. However, most studies have emphasized the phenomenon of interregional differentiation (Zhao and Lyu 2007;

Lau 2010; Chen and Zhang 2020), while the exploration of spatial network correlation features among regions has been relatively limited. The coordinated and high-quality development of regional economies does not entail uniform or homogeneous growth, but rather necessitates distinctive and mutually complementary coordinated progress. Comprehensively quantifying the quality level of China's economic development and further exploring the spatial correlation of regional high-quality development is of great significance in fully leveraging the advantages and resources of each region, improving overall economic benefits, and forming a balanced and sustainable development situation.

Therefore, based on the New Development Philosophy¹ and the concept of inclusive growth, this study conducts a comprehensive examination of the connotation of high-quality economic development, commencing from the current reality of unbalanced and inadequate economic development in China. It dynamically evaluates China's high-quality economic development level by utilizing the vertical and horizontal scatter degree approach, assigning suitable weights to six dimensions: innovation dynamic, coordinated development, market governance, green transformation, open economy, and people's welfare. Furthermore, incorporating a spatial correlation perspective, this study examines the spatial convergence of the High-Quality Economic Development Index (HQEDI) using a modified dynamic Spatial Durbin Model, revealing the multiple spatial interaction patterns of high-quality economic development across different regions and providing strong decision-making support for effectively promoting the high-quality and coordinated development of various regions in China.

2 Literature review

High-quality development has evolved into a fundamental strategy guiding the reform of China's economic system and the primary framework for innovation in development models, as the country enters in a new era of economic development. Scholars have thus conducted numerous explorations on high-quality economic development. Existing research focused mainly on three directions.

First, interpret the connotation of high-quality economic development. In a narrow sense, high-quality development refers to the efficiency of input–output production and the contribution of total factor productivity to economic growth (Yu et al. 2022). Broadly, it encompasses economic approach, structure, and dynamics that meet diverse people's needs and has multidimensional characteristics. This

¹ In 2015, the fifth plenary session of the 18th Central Committee of the Communist Party of China put forward the New Development Philosophy of innovative, coordinated, green, open and shared development, which serves as the guiding principle for China's high-quality economic development. In 2021, President Xi Jinping emphasized during a collective study session of the national leadership that innovation is the primary driving force for development, coordination is an inherent requirement for sustained and healthy development, green is a necessary condition for sustainable development and an important reflection of the people's needs for a better life, openness is the path toward national prosperity and development, and sharing is an essential requirement of socialism with Chinese characteristics.

theoretical orientation emphasizes equitable development, ecological civilization, and individual modernization (Jin 2018). The high quality of economic development is demonstrated through the intensified and stable economic growth, rationalized economic structure, and Omni-directional opening up (Feng et al. 2021; Lin and Kim 2022). The central attributes of development lie in efficiency, fairness, green and sustainability, as well as the coordination among various sectors (Zhang et al. 2019; Imran et al. 2023; Li and Yang 2023; Li et al. 2023). Supply-side reform shifts from “factor-driven” to “innovation-driven” and from “scale upgrade” to “structure upgrade” (Hua et al. 2021). Market regulation mechanisms focus on improving production factors’ quality, optimizing market allocation for efficiency, and considering non-market factors like coordination, inclusiveness, and green development (Yang and Zhang 2021). Overall, high-quality development emphasizes economic growth and coordinated development in innovation, structural optimization, market governance, open economy, ecological welfare, and social benefits.

Second, the dynamic evaluation of high-quality economic development. Within the scope of the broad concept, the dynamic evaluation of high-quality economic development requires the application of a comprehensive index system, which entails the establishment of a comprehensive set of indicators for a multidimensional evaluation, rooted in the fundamental performance of economic operations. For example, criteria developed by Nie and Jian (2019) include product and service quality, economic benefits, social benefits, ecological benefits, and favorable economic conditions. Yang and Zhang (2021) suggest that high-quality economic development can be analyzed based on economic outcome distribution, human capital distribution, economic efficiency and stability, the state of natural resources and the environment. Nevertheless, these methods are inherently subjective, thus may lead to the potential conflation of economic growth quality with that of high-quality development. Moreover, prior scholarly works have constructed assessment systems for high-quality development within the framework of the New Development Philosophy (Zhan and Cui 2016; Zhang et al. 2022). The integration of high-quality economic development objectives with policy coordination in a comprehensive measurement framework is undeniably valid. However, there is room for further enhancement in terms of the applicability of indicator selection (Li et al. 2019).

Third, the spatial evolution tendency of high-quality economic development. Previous studies primarily focus on non-spatial analysis of temporal fluctuations and regional disparities related to high-quality economic development. Conventional methodologies employed include the coefficient of variation and Theil index. However, these approaches fail to identify the existence of unique spatial correlation patterns within high-quality economic development (Zhou and Wu 2018). With the advancement of spatial econometrics, China’s scholars have employed exploratory spatial data analysis and spatial convergence models to examine the spatial disparities and convergence mechanisms of high-quality economic development (Jian and Nie 2019; Yu et al. 2022). Nevertheless, present research is largely confined to adjacent regions based on geographic distance, while in actuality, high-quality development of regions displays intricate network properties with multiple interconnected threads. Therefore, the utilization of traditional spatial econometric methods may result in biases (Liu and Jia 2019; Yu et al. 2022).

In summary, despite recent progress in the connotation interpretation, dynamic evaluation, and spatial evolution of high-quality economic development, there still remains room for further expansion in these areas. First, a unified framework for evaluating high-quality development has yet to be established. Given the development objectives and prevailing challenges within the China's economy, it is imperative to expand the depth and breadth of the connotation of high-quality development. Second, further enhancements are necessary in the exploration of high-quality development from a spatial perspective. Presently, the research overlooks the spatial correlation characteristics of the multidimensional complexity involved in high-quality development and lacks systematic studies on the spatial evolution of high-quality development concerning regional interactivity and coordination. Therefore, this study integrates the objectives of China's economic and social development with the reality of regional economic development to construct a comprehensive evaluation system for high-quality development. By employing spatial econometric models, this study reveals and explains the multiple spatial interactions and convergence patterns of the coordinated development of high-quality economies.

3 Evaluation system and research methodology

3.1 Evaluation system

3.1.1 Evaluation mechanism

China's current development is characterized by prominent imbalances and inadequacies. Specifically, these issues manifest in the inadequate adaptability of innovation capabilities to the requirements of high-quality development, significant disparities in urban–rural regional development and income distribution, considerable challenges in ecological environmental protection, shortcomings in ensuring people's welfare, and weaknesses in social governance. The requirements of China's long-range objectives of basically achieving socialist modernization include: significant improvement in innovation capabilities, advanced development of industrial foundation, significant enhancement in the coordination of urban–rural regional development, increased vitality among market entities, substantial progress in the reform of property rights system and market-oriented allocation of factors, new progress in ecological civilization construction, and achieving a new level of people's welfare. Therefore, based on the logical framework of the New Development Philosophy, the guidelines of China's 14th Five-Year Plan for economic and social development,² and in light of the reality of China's regional economy, this study constructs an evaluation

² The "Outline of the 14th Five-Year Plan for National Economic and Social Development of the People's Republic of China and the Long-Range Objectives Through the Year 2035" points out that the main goals for economic and social development during the 14th Five-Year Plan period include achieving new results in economic development, taking new steps in reform and opening up, making new advances in social civilization, achieving new progress in ecological civilization construction, reaching a new level of well-being for the people, and improving the effectiveness of national governance.

system of high-quality economic development in six dimensions, including innovation dynamic, coordinated development, market governance, green transformation, open economy, and people's welfare. The explanation is as follows.

- (1) **Innovation dynamic.** Innovation is the primary driving force to transform the economic development model from input-driven to efficiency-driven. The current extensive economic growth model in China is plagued by issues such as being large in scale but lacking strength, inadequate core innovative capabilities, and low mobility of innovative factors. To overcome the constraints of resource dependence and production patterns, it is imperative to establish a systemic environment that promotes the sharing of innovative resources and the coordination of innovation policies, thereby enhancing the efficiency of converting innovation achievements into market value creation.
- (2) **Coordinated development.** Under the persistent policy orientations toward imbalanced development, China's economic development exhibits pronounced issues of spatial resource mismatch, regional economic differentiation, and imbalanced industrial structure. Achieving high-quality development requires the resolution of accumulated contradictions in economic and social development, and the promotion of Pareto optimality within the economic system. This necessitates a focus on balancing spatial distribution and enhancing market efficiency, in order to facilitate the optimized allocation of production factors between urban and rural areas as well as among different regions.
- (3) **Market regulation.** The primary challenge of augmenting the efficacy of the market mechanism as a catalyst for high-quality economic development lies in fully exploiting the decisive role of the market in resource allocation. This entails reducing government intervention, lowering government consumption and investment, decreasing the proportion of monopolistic enterprises, and increasing investment and output from non-public sector entities. Additionally, it requires strengthening the fundamental role of the market in facilitating the mobility of labor, capital, and other production factors, while cultivating the vitality of diverse market participants.
- (4) **Green transformation.** Integrating material capital with natural capital to resolve the contradiction between economic development and environmental damage is a crucial pathway for constructing high-quality development. To promote the coordinated advancement of ecological civilization construction and economic growth transformation, and to achieve an economically high-quality development model that optimizes the value of natural capital, it is essential to prioritize the protection of ecological environment welfare. This requires the establishment of an evaluation system for the goals of ecological civilization construction, along with enhancing the governance of regulatory indicators such as environmental protection, natural resource management, energy conservation, and emission reduction (Imran and Zhang 2023; Li et al. 2023).
- (5) **Open economy.** In the era of economic globalization, open development entails aligning with the current trend, proactively integrating into the international value chain division of labor and leveraging advanced technological advancements and management experience to enhance domestic competitiveness. It also

entails actively engaging in global economic governance to advance globalization in a manner that is characterized by transparency, inclusivity, balance, and mutual benefit. Therefore, open development should be guided by equal emphasis on both “bringing in” and “going global,” strengthening the efficiency of foreign capital utilization, and enhancing the quality of external cooperation to deepen engagement in the global value chain.

- (6) **People’s welfare.** Throughout the period of China’s rapid economic growth, the prioritization of efficiency has, to some extent, compromised the well-being of individuals, leading to disparities in income and wealth, as well as insufficient provisions for cultural education and social welfare. As demographic dividends diminish and marginal returns of input factors decline, promoting equitable and inclusive development becomes pivotal for achieving high-quality economic development. Consequently, key goals in implementing shared development include restructuring income distribution, promoting widespread access to cultural education, ensuring basic livelihoods for the disadvantaged, and enhancing social welfare provisions.

3.1.2 Indicator system

This study further establishes 15 objectives, which are based on the six main dimensions and aligned with the primary goals of China’s economic and social development throughout the 14th Five-Year Plan period: innovation inputs, innovation efficiency, urban–rural coordination, industrial coordination, regional coordination, supply–demand balance, market vitality, pollution emission, resource consumption, environmental regulation, level of openness, openness effect, poverty alleviation, income distribution and social welfare. Following the consolidation of 56 indicators reflecting China’s real economic and social progress, the dynamic evaluation system for high-quality economic development is formulated. Table 1 lists the selected indicators, the computation method, and the reference base.

3.1.3 Data source

In this study, data from 2002 to 2019 for 30 provinces, municipalities, and autonomous regions (in the subsequent text, these administrative areas are collectively referred to as “provinces”)—aside from Tibet, Hong Kong, Macao, and Taiwan—were chosen from the “China Statistical Yearbook,” the “China Science and Technology Statistical Yearbook,” the “China Environmental Statistical Yearbook,” the “China Energy Statistical Yearbook,” as well as the official websites of the provincial statistical bureaus. Where CO₂ emissions are quantified in accordance with the guidelines in the “National Greenhouse Gas Emissions Inventory (2006 edition).”

Table 1 Indicator system for high-quality economic development in China

Dimension	Objectives	Indicators	Indicator computation methods	Unit	Attributes	References
Innovation dynamic	Innovation inputs	R&D input intensity	R&D investment/local financial expenditure	%	+	Wan et al. (2023)
		R&D investment efforts	R&D employees/local employment	%	+	
		Technology support intensity	Technology appropriation/local financial expenditure	%	+	Chen et al. (2020a, b)
		Industry regulation level	High-tech industry investment/GDP	%	+	Li et al. (2019)
	Innovation efficiency	Higher education level	Population with graduate degree/Total population	%	+	Shi and Zhang (2019)
		Domestic patents granted	Number of domestic patents granted/total population	Pieces/10,000 people	+	Yu et al. (2022)
		Technology market turnover	Technology market turnover/GDP	%	+	Wei and Li (2018)
		New product sales revenue	Sales revenue of new high-tech products/GDP	%	+	Xu et al. (2021)

Table 1 (continued)

Dimension	Objectives	Indicators	Indicator computation methods	Unit	Attributes	References
Coordinated development	Urban–rural coordination	Binary comparison coefficient	Agricultural comparative labor productivity/Non-agricultural comparative labor productivity	%	+	Zhan and Cui (2016)
		Binary contrast coefficient	Share of non-agricultural output—Share of non-agricultural employment	%	–	
		Gini coefficient	Gini coefficient	%	–	Li et al. (2019)
		Urban and rural consumption levels	Ratio of urban and rural per capita consumption levels	%	–	Wan et al. (2023)
	Industrial coordination	Advanced industrial structure	Industrial structure advanced index	%	+	Gan et al. (2011)
		Primary industry labor productivity	Comparative labor rate of primary industry	%	+	
		Secondary industry labor productivity	Comparative labor rate of secondary industry	%	+	Li and Ren (2019)
		Tertiary industry labor productivity	Comparative labor rate of tertiary industry	%	+	
		Rate of product superiority	Product quality superiority rate	%	+	Zhu (2019)
	Regional coordination	Regional income level	Regional GDP per capita/national GDP per capita	%	+	Xu and Ding (2019)
		Regional consumption level	Regional consumption level/national per capita consumption	%	+	

Table 1 (continued)

Dimension	Objectives	Indicators	Indicator computation methods	Unit	Attributes	References
Market regulation	Supply–demand balance	Social employment level	Social employment rate	%	+	Wei and Li (2018); Jian and Nie (2019)
		Price volatility	Inflation rate	%	–	
		Economic volatility	Volatility of economic growth	%	–	
	Market vitality	Market dominance level	Non-governmental investment/total investment	%	+	Wei and Ren (2012); Wei and Li (2018)
		Investment diversification level	Non-state investment/total investment	%	+	
		Marketization of capital elements	Financial value-added/GDP	%	+	
		Marketization of labor factors	Number of private sector employment/total regional employment	%	+	
Green transformation	Pollution emission	Wastewater pollution	Total wastewater discharge/GDP	Tons/million CNY	–	National Development and Reform Commission, Department of Environmental Resources [2016] No. 2635 ^a
		Exhaust gas pollution	Sulfur dioxide emissions/GDP	Tons/million CNY	–	
		Solid waste pollution	General industrial fixed waste emissions/GDP	Tons/million CNY	–	
		Carbon emission level	Carbon dioxide emissions/GDP	Tons/million CNY	–	
	Resource consumption	Energy utilization efficiency	GDP/ton of standard coal	Tons/million CNY	+	Chen et al. (2022)
		Electricity utilization efficiency	GDP/electricity consumption	Kilowatt/million CNY	+	

Table 1 (continued)

Dimension	Objectives	Indicators	Indicator computation methods	Unit	Attributes	References
Open economy	Environmental regulation	Environmental investment level	Environmental treatment investment/Total local investment	%	+	National Development and Reform Commission, Department of Environmental Resources [2016] No. 2635 ^b
		Solid waste treatment	General fixed waste comprehensive utilization rate	%	+	
		Exhaust gas treatment level	Sulfur dioxide removal rate	%	+	
		Wastewater treatment level	Municipal wastewater discharge compliance rate	%	+	
		Urban greening level	Greening coverage rate of urban built-up areas	%	+	
	Level of openness	Net exports of goods and services	Net exports of goods and services/GDP	%	+	Tao and Xu (2021)
		Laborers to foreign countries	Number of workers working abroad/total population	%	+	
		High-tech industry trade	High-tech industry import–export trade/GDP	%	+	
		Foreign trade dependence	Total imports and exports/GDP	%	+	

Table 1 (continued)

Dimension	Objectives	Indicators	Indicator computation methods	Unit	Attributes	References
People's welfare	Opening effect	Scale of foreign economic cooperation	Foreign economic cooperation turnover/GDP	%	+	Chen et al. (2020a, b)
		Frequency of foreign economic cooperation	Number of foreign economic cooperation contracts	Number	+	Lyu and Cui (2020) Li et al. (2022)
		Foreign investment dependence	Actual utilization of foreign capital/GDP	%	+	Xu et al. (2020)
		Internationalization level of tourism	International tourism foreign exchange earnings/GDP	%	+	Yang (2019)
	Poverty alleviation	Share of poor population	Population receiving the minimum guarantee/total population of the region	%	–	“Regional Development and Livelihood Index” National Bureau of Statistics
		Size of social security fund	Social security fund revenue/GDP	%	+	
		Level of initial income Distribution	Labor compensation/GDP	%	+	Martinez-Vazquez et al. (2012)
	Income distribution	level of social income redistribution	Social security and employment expenditure/GDP	%	+	

Table 1 (continued)

Dimension	Objectives	Indicators	Indicator computation methods	Unit	Attributes	References
	Social welfare	Cultural facilities services	Public book collection per capita	Number of books/person	+	Pan et al. (2021)
		Level of medical self-sufficiency	Beds per capita in medical institutions	Number of beds/person	+	
		Internet penetration rate	Households with internet broadband connections/total households	%	+	
		Level of basic education	Years of schooling per capita	Years	+	Fang and Ma (2019)
		population mortality rate	Population mortality rate	%	–	
		Tourism Engel's coefficient	Engel's coefficient of resident tourism	%	+	Sun and Yang (2014)
		Transportation facilities	Total road miles per 10,000 people	Kilometers/10,000 people	+	Zhou and Wu (2018)

"+" in the attribute indicates that the indicator is positive, "-" indicates that the indicator is negative.

^aDepartment of Environmental Resources, Assessment System for Ecological Civilization Construction (2016). General Office, 2635. Retrieved from https://www.ndrc.gov.cn/fggz/hjzy/stwnjs/201612/20161222_1161174_ext.html.

^bDepartment of Environmental Resources, The Green Development Indicator System (2016). General Office, 2635. Retrieved from https://www.ndrc.gov.cn/fggz/hjzy/stwnjs/201612/20161222_1161174_ext.html.

3.2 Research methodology

3.2.1 Dynamic evaluation

This study introduces measures to rectify the shortcomings of past studies and ensure objectivity, dynamic comparability, as well as transparency and repeatability in the evaluation process. The raw data are normalized using the efficacy coefficient base-setting approach, followed by the application of the vertical and horizontal scatter degree method to weight the basic indicators. Subsequently, the provincial HQEDI is calculated using the linear weighting method.

- (1) Efficacy coefficient base-setting approach. This approach standardizes the raw data using 2002 as the base year, to make the indexes comparable across different years. The precise formulas are:

$$s_{ij}(t_k) = \begin{cases} \frac{\max[x_j(t_0)] - x_{ij}(t_k)}{\max[x_j(t_0)] - \min[x_j(t_0)]} & x_j \text{ is the inverse indicator} \\ \frac{x_{ij}(t_k) - \min[x_j(t_0)]}{\max[x_j(t_0)] - \min[x_j(t_0)]} & x_j \text{ is the positive indicator} \end{cases} \quad (1)$$

where $x_{ij}(t_k)$ and $s_{ij}(t_k)$ denote the original and normalized values of the j th indicator for the i th province in year t_k . $\max[x_j(t_0)]$ and $\min[x_j(t_0)]$ denote the maximal and minimum values of the j th indicator in the base period t_0 for all provinces.

- (2) Vertical and horizontal scatter degree method. Setting the object collection to be evaluated as $s = \{s_1, s_2, \dots, s_n\}$, the vector of weighting coefficients as $w = (w_1, w_2, \dots, w_n)^T$, and let $x_{ij}(t_k)$ be the original data of the j th index in province i in year t_k , which are changed to $x_{ij}^*(t_k)$ by dimensionless processing of extreme values, and the evaluation function in year t_k is $y_i(t_k) = \sum_{j=1}^m w_j x_{ij}^*(t_k)$.

The principle of determining the weight coefficients is to maximize the differences between the evaluation objects in the time-series stereo data, which is portrayed by the total sum of squared deviations of $y_i(t_k)$. This weighting method, in comparison with commonly used Entropy-weighting or Subjective-weighting methods in existing research, exhibits significant precision, particularly in studies of regional disparities, where it serves to accentuate differences with greater clarity.

$$\delta^2 = \sum_{k=1}^N \sum_{i=1}^n [y_i(t_k) - \bar{y}]^2 \quad (2)$$

$$\bar{y} = \frac{1}{N} \sum_{k=1}^n \left(\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^m \omega_j x_{ij}(t_k) \right) = 0 \quad (3)$$

The essence of extremum dimensionless processing is to standardize the original data so that the total sum of squares of the deviations can be simplified as:

$$\delta^2 = \sum_{k=1}^N \sum_{i=1}^n [y_i(t_k)]^2 = \sum_{k=1}^N [w^T H_k w] = w^T \sum_{k=1}^N [H_k] w = w^T H w \quad (4)$$

where H is a symmetric matrix of order $m \times m$, $H_k = A_k^T A_k$

$$A_k = \begin{Bmatrix} x_{11}(t_k) & \dots & x_{1m}(t_k) \\ \vdots & \ddots & \vdots \\ x_{n1}(t_k) & \dots & x_{nm}(t_k) \end{Bmatrix}, k = 1, 2, \dots, N \quad (5)$$

Further, setting $w^T w = 1$, then, the value of w at this point is the maximum eigenvector corresponding to the maximum eigenvalue of the matrix H , and the value of δ^2 is the largest. To ensure that all coefficients are positive, setting $w > 0$, the eigenvectors of weights are obtained by solving the following linear programming problem.

$$\max \quad w^T H w \quad s.t. \|w\| = 1 \quad \text{and} \quad w > 0 \quad (6)$$

- (3) HQEDI measurement. By combining the standardized index values and the vector of weighting coefficients, the measurement first apply the linear weighting method to find the HQEDI value $QDI_i(t_k)$ of the i th region in year t_k , then use the real GDP of each province in the country (2002 as the base year) as the weighting coefficient, and then apply the linear weighting method to convert the index $QDI_i(t_k)$ of each province into the total index $QDI_r(t_k)$ of the r th region in year t_k .

$$QDI_i(t_k) = \sum_{j=1}^m w_j s_{ij}(t_k) \quad (7)$$

$$QDI_r(t_k) = \sum_{i=1}^q (y_i^t / Y_r^t) I_i(t_k) \quad (8)$$

where q is the number of provinces in region r , y_i^t denotes the output y of province i in period t , and Y_r^t denotes the total output Y of region r in period t . The significant variations in the economic scale of China's provinces suggest unequal positions within the regional economy. Utilizing a linear weighting method to calculate the aggregate index is superior to direct summation, as it better captures the nuanced characteristics of China's regional economy by accounting for the diverse impact of each province.

3.2.2 Spatial correlation

The spatial correlation of high-quality economic development refers to a network of high-quality development interactions that occur between regions (Scott and Carrington 2014; Liu and Jia 2019). To assess the influence of economic and geographical distances on the spatial correlation of high-quality regional development and to provide a more precise depiction of the spatial correlation status of regional

economic high-quality development, this study made appropriate adjustments to the gravity model (Lundmark 2017), as is displayed in Eq. (9). Where X_{ij} denotes the special gravity force of high-quality economic development between province i and j ; Q_i and Q_j , P_i and P_j , G_i and G_j are the HQEDI, the resident population and the real regional GDP of provinces i and j , respectively; λ_{ij} is the contribution of province i in the high-quality economic development interaction between provinces i and j ; D_{ij} is the distance between the provinces' capitals i and j ; the per capita difference in gross output value $(g_i - g_j)$ denotes the economic geographic distance between provinces.

On the basis of the modified gravity model, the gravity matrix for the development of regional economic quality is constructed. This study uses the mean value of each row of the gravity matrix as the critical value. The gravity value higher than the critical value of the row is recorded as 1, which indicates that there is a spatial correlation between the provinces in the row and the provinces in the column for high-quality economic development; the gravity value lower than the critical value of the row is recorded as 0, which indicates that there is no spatial correlation. The gravity matrix can then be converted into a relationship matrix that reflects spatial correlation in this manner.

$$X_{ij} = \lambda_{ij} \frac{\sqrt[3]{P_i Q_i G_i} \sqrt[3]{P_j Q_j G_j}}{[D_{ij} / (g_i - g_j)]^2} \quad (9)$$

3.2.3 Spatial convergence

Drawing on the concept of β convergence in neoclassical economic growth theory, this study conducts estimations to analyze the spatial convergence of China's regional high-quality economic development level. β convergence can be classified into absolute convergence and conditional convergence, with the conditional convergence aligns more closely with economic reality by assuming the presence of interregional variability, diverse economic characteristics, and distinct steady-state values across different regions. Therefore, the focus of this study lies in the examination of conditional convergence. The spatial weights matrix consists of adjacent distance weights, geographic distance weights, economic distance weights, geographic-economic distance weights, and spatial correlation weights. The model is specified as follows in the dynamic Spatial Durbin panel model introduced by LeSage and Pace (2009):

$$\begin{aligned} \text{Ln} \left(\frac{\text{QDI}_{i,t+1}}{\text{QDI}_{i,t}} \right) = & \lambda \sum_{j=1}^n W_{ij} \text{Ln} \left(\frac{\text{QDI}_{i,t+1}}{\text{QDI}_{i,t}} \right) + \beta \text{LnQDI}_{i,t} + \phi_n X_{i,t} \\ & + \eta \sum_{j=1}^n W_{ij} \text{LnQDI}_{i,t} + \mu \sum_{j=1}^n W_{ij} X_{i,t} + u_i + v_i + \varepsilon_{it} \end{aligned} \quad (10)$$

where λ is the lagged spatial regression coefficient, W_{ij} is the element of the i th row and j th column of the standardized spatial weights matrix, X_{it} is the set of control

Table 2 Descriptive statistics of control variables

Variables	Average	Standard deviation	Minimum value	Maximum value	Observation volume
EOL	0.311	0.381	0.017	1.722	510
ETL	1.187	0.744	0.290	5.140	510
IDL	0.386	0.085	0.118	0.592	510
LWAGE	4.510	0.272	3.963	5.120	510
ML	0.262	0.161	0.040	0.982	510
PGL	0.003	0.001	0.001	0.014	510

variables, u_i and v_i denote the time and spatial effects, respectively, and ε_{it} is the error term. If the model converges to the same long-run equilibrium after adding the control variables, that is, β is significantly < 0 , this implies the existence of conditional convergence during the sample period.

The regional openness level, marketization level, per capita living standard, resource transformation level, industrial development level, and government governance level are chosen as the control variables based on the findings of the previous studies. Among them, regional openness level (EOL) is expressed by the proportion of total import and export to GDP, marketization level (ML) is expressed by the proportion of budgeted fiscal expenditure to GDP, per capita living standard (LWAGE) is expressed by the logarithm of average wage of employees, resource transformation level (ETL) is expressed by the rate of decrease in energy consumption per unit, industrial development level (IDL) is expressed by the proportion of value added of tertiary industry to the value added of secondary industry, and government governance level (PGL) is expressed as the number of cases of official crimes filed per 10,000 public officials. All data were obtained from the “China Statistical Yearbook,” “China Energy Statistical Yearbook,” “China Procuratorial Yearbook,” and the reports of the People’s Procuratorates in each province. The descriptive statistics of the control variables are shown in Table 2.

4 Dynamic evaluation of China’s high-quality economic development

4.1 National level

According to Table 3, the national HQEDI in China showed a steady improvement from 0.501 in 2002 to 0.616 in 2019, with an average annual growth rate of 1.36%. In terms of the fluctuation cycles of economic and industrial growth, the time span of high-quality development in China has gradually lengthened. China’s total GDP increased from RMB 12.17 trillion in 2002 to RMB 98.69 trillion in 2019, with an average annual growth rate of 13.10%. The industrial value-added increased from RMB 4.78 trillion in 2002 to RMB 31.19 trillion in 2019, with an average annual growth rate of 11.67%. The service value-added industry increased from RMB 4.57

Table 3 China's HQEDI by Provinces and Four Regions

Region	2002	2004	2006	2008	2010	2012	2014	2016	2019	Mean	Rank	Growth Rate %	Rank
Beijing	0.612	0.629	0.680	0.705	0.724	0.717	0.737	0.739	0.778	0.706	1	1.42	20
Tianjin	0.595	0.615	0.625	0.639	0.641	0.646	0.670	0.674	0.711	0.645	3	1.05	28
Hebei	0.465	0.460	0.490	0.516	0.536	0.526	0.554	0.570	0.611	0.527	18	1.62	16
Shanxi	0.380	0.407	0.437	0.471	0.500	0.506	0.528	0.546	0.596	0.489	24	2.68	3
Inner Mongolia	0.399	0.404	0.445	0.485	0.502	0.513	0.542	0.544	0.569	0.491	23	2.11	8
Liaoning	0.513	0.528	0.545	0.567	0.591	0.595	0.608	0.586	0.627	0.573	9	1.19	27
Jilin	0.500	0.479	0.521	0.510	0.545	0.563	0.576	0.598	0.634	0.549	14	1.41	21
Heilongjiang	0.462	0.471	0.498	0.500	0.531	0.533	0.532	0.570	0.606	0.525	19	1.61	17
Shanghai	0.636	0.639	0.665	0.667	0.696	0.688	0.714	0.712	0.739	0.684	2	0.89	30
Jiangsu	0.551	0.552	0.587	0.604	0.623	0.630	0.649	0.648	0.687	0.612	6	1.31	24
Zhejiang	0.534	0.553	0.580	0.593	0.610	0.623	0.649	0.660	0.692	0.616	5	1.54	18
Anhui	0.475	0.483	0.499	0.528	0.562	0.573	0.592	0.619	0.645	0.556	11	1.82	13
Fujian	0.529	0.542	0.542	0.553	0.583	0.583	0.614	0.621	0.652	0.582	8	1.24	26
Jiangxi	0.466	0.474	0.497	0.530	0.545	0.567	0.590	0.603	0.640	0.550	13	1.89	12
Shandong	0.522	0.528	0.556	0.573	0.593	0.590	0.616	0.607	0.650	0.583	7	1.30	25
Henan	0.451	0.460	0.486	0.514	0.536	0.547	0.574	0.586	0.624	0.533	17	1.93	10
Hubei	0.469	0.473	0.495	0.521	0.555	0.568	0.595	0.612	0.646	0.553	12	1.90	11
Hunan	0.466	0.465	0.493	0.513	0.542	0.545	0.569	0.597	0.626	0.539	15	1.75	15
Guangdong	0.560	0.553	0.594	0.612	0.627	0.616	0.655	0.670	0.702	0.623	4	1.34	22
Guangxi	0.416	0.415	0.451	0.465	0.497	0.503	0.534	0.557	0.595	0.496	22	2.13	7
Hainan	0.520	0.524	0.536	0.567	0.582	0.573	0.578	0.583	0.617	0.563	10	1.01	29
Chongqing	0.411	0.424	0.454	0.491	0.540	0.544	0.567	0.594	0.634	0.521	20	2.58	5
Sichuan	0.449	0.450	0.478	0.517	0.538	0.560	0.588	0.603	0.627	0.539	16	1.98	9

Table 3 (continued)

Region	2002	2004	2006	2008	2010	2012	2014	2016	2019	Mean	Rank	Growth Rate %	Rank
Guizhou	0.290	0.311	0.344	0.401	0.455	0.485	0.511	0.539	0.574	0.438	29	4.10	1
Yunnan	0.410	0.428	0.435	0.460	0.479	0.488	0.496	0.509	0.514	0.470	26	1.34	23
Shaanxi	0.405	0.412	0.457	0.473	0.505	0.522	0.564	0.580	0.604	0.507	21	2.38	6
Gansu	0.360	0.377	0.402	0.428	0.451	0.471	0.501	0.522	0.562	0.458	27	2.65	4
Qinghai	0.346	0.361	0.379	0.400	0.413	0.433	0.429	0.433	0.469	0.407	30	1.81	14
Ningxia	0.354	0.359	0.382	0.431	0.459	0.482	0.503	0.519	0.556	0.450	28	2.69	2
Xinjiang	0.423	0.441	0.443	0.472	0.487	0.511	0.518	0.523	0.542	0.488	25	1.47	19
The Eastern Region	0.549	0.553	0.584	0.600	0.619	0.617	0.644	0.649	0.663	0.612	1	1.12	4
The Central Region	0.455	0.463	0.487	0.514	0.542	0.553	0.577	0.597	0.672	0.542	3	2.32	1
The Western Region	0.407	0.416	0.444	0.476	0.503	0.519	0.545	0.560	0.581	0.499	4	2.12	2
The Northeastern Region	0.494	0.500	0.525	0.535	0.563	0.570	0.579	0.584	0.613	0.554	2	1.28	3
National Average	0.501	0.508	0.537	0.557	0.579	0.583	0.607	0.616	0.630	0.572	—	1.36	—

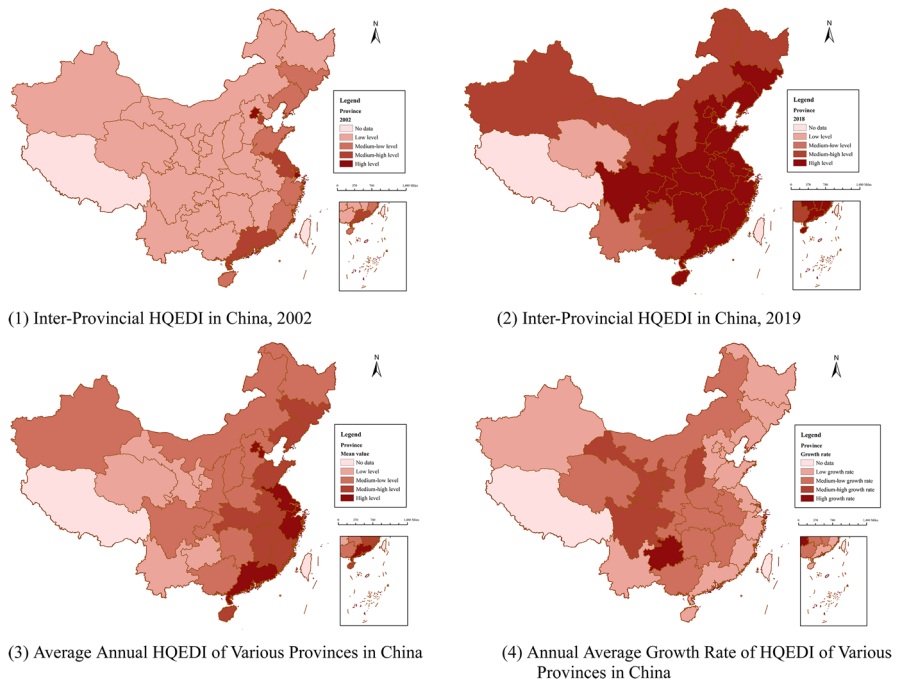


Fig. 1 China's HQEDI by provinces

trillion in 2002 to RMB 53.42 trillion in 2019, with an average annual growth rate of 15.56%.

From this perspective, China's economic growth rate exhibits stability, with consistent industrial expansion and a heightened pace of adjustment. These factors serve as a quantitative basis for the intensity, stability, and rationality inherent in achieving high-quality development. China has embarked on a trajectory of synergistic enhancement, wherein stable economic growth and high-quality development reciprocally reinforce each other.

4.2 Regional level

From Table 3, it can be observed that the HQEDI in the eastern region has consistently outperformed the central, western and northeastern regions, increasing from 0.549 in 2002 to 0.663 in 2019, with an average annual growth rate of 1.12%. The HQEDI in the northeastern region increased from 0.494 in 2002 to 0.613 in 2019, with an average annual growth rate of 1.28%. The central region demonstrates a relatively lower HQEDI but the fastest growth rate, increasing from 0.455 in 2002 to 0.672 in 2019, with an average annual growth rate of 2.32%. The western region exhibits the lowest HQEDI but a higher growth rate, increasing from 0.407 in 2002

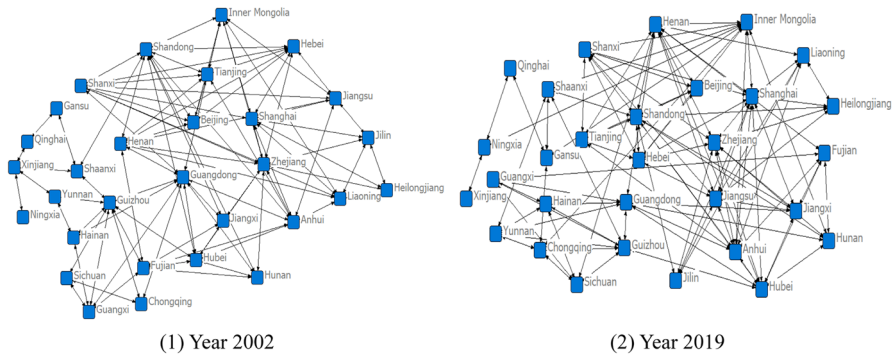


Fig. 2 Spatial correlation of China's high-quality economic development in 2002 and 2019

to 0.581 in 2019, with an average annual growth rate of 2.12%. Figure 1 visually illustrates the disparities among provinces and regions in terms of HQEDI.

It can be inferred that the spatial distribution of the HQEDI in China exhibits significant regional disparities, showing a gradual decrease from the eastern to the western regions, while the growth rate demonstrates a gradual acceleration in the same direction. This indicates that as reforms have advanced, the central and western regions have unleashed their development potential, leading to a “catch-up effect” toward the eastern region and a discernible reduction in regional disparities (Jian and Nie 2019). However, it is notable that the northeastern region has undergone significant “collapse,” with the development gap widening from 0.055 in 2002 to 0.071 in 2018 (this number experienced a severe downward fluctuation between 2018 and 2019) compared to the eastern region, and from 0.039 in 2002 to 0.059 in 2019 compared to the central region. The northeastern region is currently facing the dual constraints of low economic growth rate and development quality.

5 Spatial evolution of China's high-quality economic development

5.1 Spatial correlation features

This study utilizes a modified gravity model to construct a spatial correlation matrix for high-quality regional economic development among China's provinces. Figure 2 illustrates the spatial correlation network, highlighting three significant characteristics: First, the spatial correlation network illustrates variations in regional status, with Shanghai and Beijing acting as primary centers, while other eastern provinces and cities function as secondary centers, with central and western provinces occupying a subordinate position in the network. Second, within spatially correlated networks, interwoven channels facilitate the transmission of mutual influence, suggesting the presence of intimate spatial correlation relationships among provinces that surpass conventional geographical adjacency. Thirdly, spatially correlated networks

present holistic attributes that advance high-quality regional economic development, where such progress is the result of collective efforts from all regions in China.

5.2 Spatial convergence test

To visually demonstrate the spatial convergence patterns of high-quality development in China within the framework of spatial correlation, and to enhance the robustness of the findings, this study calculates the spatial autocorrelation of HQEDI in China using Moran's I index with consideration given to adjacent distance weights, geographic distance weights, economic distance weights, geographic-economic distance weights, and spatial correlation weights. Results in Table 4 indicate significant spatial correlations of HQEDI under different spatial weights.

The Hausman test results support the fixed effects model and combined with the results of the Wald and LR tests, it can be concluded that the dynamic Spatial Durbin Model with lagged terms of explanatory variables is a superior choice. Finally, considering the potential endogeneity of the model, this study employs the maximum likelihood estimation (MLE) method to eliminate estimation bias. The results are displayed in Table 5.

Under no weight condition, the estimated coefficients β for the HQEDI are significantly negative, and a stochastic convergence impact is present with a convergence rate of 1.78% and half-way convergence cycle of 39.22 years. For various spatial weights, the HQEDI exhibits conditional convergence when the spatial element is taken into account, with a significantly positive spillover impact. Specifically, the convergence rate of the HQEDI under the adjacent distance weights is 1.72%, and half-way convergence cycle is 40.68 years; the numbers are 1.53% and 45.54 years under the geographic distance weights, 1.67% and 41.91 years under the economic distance weights, 1.33% and 52.64 years under the geographic-economic distance weights, 1.64% and 42.61 years under the spatial correlation weights. Based on the analysis, several observations can be made. First, the introduction of spatial weights leads to a diverse decrease in the convergence rate of the HQEDI, coupled with an extension of the convergence cycle. This indicates that the spatial correlation patterns impose constraints on the convergence rate and duration of high-quality development due to the inherent path heterogeneity and complexity. Second, the incorporation of economic distance and spatial correlation weights results in an accelerated convergence rate of the HQEDI and a shortened convergence cycle. This finding suggests a resemblance in the spatial interaction patterns between regional economic scale and high-quality development. Specifically, as the economic distance between regions decreases, the spatial correlation becomes stronger, thereby further expediting the convergence rate. Third, the slowest convergence rate and longest convergence cycle are observed when incorporating geographic-economic distance weights. This outcome can be attributed to the distribution pattern of regional economic development disparities, characterized by a "center-periphery" model, thereby leading to polarization and diffusion effects. The diffusion effect from more developed regions supports the improvement of high-quality development levels in adjacent regions through mechanisms such as resource transfer and factor diffusion.

Table 4 Global Moran's I Index of China's Regional HQEDI

Year	Adjacent distance weights		Geographic distance weights		Economic distance weights		Geographic-economic distance weights		Spatial correlation weights	
	Moran's I value	z value	Moran's I value	z value	Moran's I value	z value	Moran's I value	z value	Moran's I value	z value
2002	0.541***	4.677	0.422***	4.779	0.353***	3.997	0.033**	1.988	0.242***	2.797
2003	0.521***	4.514	0.430***	4.869	0.338***	3.844	0.028*	1.838	0.267***	3.058
2004	0.542***	4.682	0.442***	4.990	0.361***	4.086	0.032**	1.962	0.232***	2.695
2005	0.536***	4.650	0.444***	5.025	0.375***	4.236	0.031*	1.900	0.241***	2.793
2006	0.537***	4.661	0.441***	5.002	0.389***	4.386	0.030*	1.893	0.250***	2.891
2007	0.530***	4.590	0.428***	4.844	0.382***	4.302	0.032*	1.950	0.232***	2.705
2008	0.538***	4.683	0.440***	5.002	0.407***	4.581	0.034**	1.984	0.209***	2.482
2009	0.509***	4.459	0.423***	4.842	0.413***	4.665	0.033**	1.964	0.223***	2.636
2010	0.528***	4.620	0.435***	4.966	0.401***	4.539	0.031*	1.896	0.245***	2.857
2011	0.478***	4.202	0.433***	4.947	0.422***	4.754	0.035**	2.004	0.209***	2.488
2012	0.491***	4.314	0.451***	5.136	0.418***	4.716	0.035**	2.023	0.214***	2.544
2013	0.483***	4.232	0.433***	4.929	0.395***	4.466	0.029*	1.844	0.224***	2.635
2014	0.464***	4.099	0.428***	4.894	0.417***	4.715	0.034**	1.978	0.190**	2.297
2015	0.467***	4.128	0.433***	4.956	0.378***	4.309	0.030*	1.872	0.273***	3.157
2016	0.461***	4.100	0.419***	4.833	0.352***	4.064	0.030*	1.857	0.237***	2.803
2017	0.444***	3.970	0.383***	4.462	0.328***	3.820	0.028*	1.796	0.240***	2.836
2018	0.447***	3.991	0.376***	4.388	0.316***	3.693	0.028*	1.792	0.238***	2.815
2019	0.485***	4.240	0.402***	4.589	0.323***	3.709	0.051***	4.553	0.273***	3.128

*, **, *** indicate significant at the 10%, 5%, and 1% levels, respectively

Table 5 Spatial convergence estimation results of China's HQEDI

Variables	No weight	Adjacent distance weights	Geographic distance weights	Economic distance weights	Geographic-economic distance weights	Spatial correlation weights
LnQDI	−0.2347*** (−7.20)	−0.2272*** (−5.59)	−0.2055*** (−5.14)	−0.2213*** (−5.53)	−0.1803*** (−4.41)	−0.2181*** (−5.46)
EOL	−0.0069** (−2.06)	−0.0005 (−0.13)	−0.0005 (−0.13)	−0.0210** (−2.23)	−0.0231 (−0.29)	−0.0026 (−0.70)
ML	0.0168*** (3.63)	0.0173** (2.31)	0.0153* (1.95)	0.0051 (0.53)	0.0079 (0.69)	0.0232*** (2.62)
LWAGE	−0.0133*** (−5.32)	−0.0021 (−0.14)	−0.0001 (−0.12)	−0.0086 (−0.56)	−0.0010 (−0.06)	−0.0003 (−0.18)
ETL	0.0054*** (7.23)	0.0048*** (3.20)	0.0046*** (3.17)	0.0059*** (4.33)	0.0053*** (3.69)	0.0043*** (3.21)
IDL	0.0081*** (3.56)	0.0011 (0.58)	0.0061 (0.32)	0.0034 (1.01)	0.0004 (0.19)	0.0005 (0.26)
PGL	−4.9955*** (−4.18)	−1.1091** (−2.40)	−2.9466* (−1.71)	−5.3375*** (−3.34)	−3.9804** (−1.98)	−1.1401* (−1.74)
Constant Term	0.0667*** (4.89)					
λ		0.3839*** (7.84)	0.4757*** (9.18)	0.4427*** (8.26)	0.5378*** (7.61)	0.4521*** (9.12)
Convergence	Yes	Yes	Yes	Yes	Yes	Yes
Rate	0.0178	0.0172	0.0153	0.0167	0.0133	0.0164
logL		1583.91	1593.37	1596.77	1590.82	1597.15
AIC		−3137.81	−3156.73	−3163.53	−3151.64	−3164.30
BIC		−3076.18	−3095.09	−3101.89	−3089.99	−3102.67

*, **, *** indicate significant at the 10%, 5%, and 1% levels, respectively

However, the existence of developmental disparities between regions hinders the developmental progress of neighboring regions through a siphon effect, thereby reinforcing the polarization trends (Yang and Li 2020).

Of the control variables examined, the regional openness level demonstrates a negative influence and attains statistical significance solely under conditions of no weight and economic distance weights. This suggests that variations in openness level will impede the pace of high-quality economic development and exacerbate regional development disparities. Conversely, the marketization level demonstrates a positive effect and attains statistical significance, except when considering economic distance and spatial correlation weights. This implies that increased government fiscal spending can expedite the rate of high-quality economic development and facilitate its mutually beneficial enhancement. The per capita living standard exerts negative impact, showing significance only when no weight is applied. This indicates that worker income levels hinder convergence, but the lack of significance under

spatial factors implies that population mobility may partially offset this impact. The resource transformation level and industrial development level have positive impacts, suggesting that the acceleration of the decline in per unit energy consumption and the improvement of industrial structure can enhance inter-provincial economic development synergy. Government governance negatively impacts development, as evidenced by the increase in registered cases of official crimes leading to lower operational efficiency of local governments, consequently resulting in a slower progression toward high-quality development.

5.3 Club convergence test

Considering the disparity in high-quality development across the four major regions in China, this study further explores the impact of spatial effects on the convergence of regional HQEDI. The Club Convergence trends of high-quality development across various regional levels are re-evaluated using geographic-economic distance and spatial correlation weight matrices. Table 6 shows the spatial convergence estimation results of the regional HQEDI.

The estimated coefficients β are negative in the eastern, central, western, and northeastern regions under the weights of geographic-economic distance and spatial correlation, all of which demonstrate significant spatial convergence effects. Specifically, under the weight of geographical and economic distance, the convergence rates for the eastern, central, western, and northeastern regions are 1.0%, 2.8%, 1.8%, and 1.6%, respectively. The corresponding half-way convergence cycles are 68.6 years, 24.8 years, 39.2 years, and 42.9 years. Convergence rates under spatial correlation weighting were 1.0%, 2.4%, 2.1%, and 0.9% for the eastern, central, western, and northeastern regions, respectively, corresponding to half-way convergence cycles of 71.7, 29.2, 33.5, and 78.8 years, respectively. Drawing from these observations, it can be inferred that the convergence rate of regional HQEDI experiences a decline in the direction of Central-West-Northeast-East, while simultaneously witnessing an increase in the convergence cycle within the same direction. This signifies that the siphon effect is leading to the migration of factors such as talent, capital, and technology from central and western China toward the East, consequently diminishing the disparities in factor endowments among the central and western provinces. As a result, there is an observable trend toward the gradual convergence of HQEDI growth rates in the central and western regions. The convergence rate in the eastern region is comparatively slower when contrasted with that of the central and western regions. This suggests that the polarization effect resulting from the interregional flow of factors is more pronounced in the eastern region under the influence of spatial factors, thereby intensifying the development disparity within the region. Under the spatial correlation weighting, the convergence rates in the eastern, central, and northeastern regions experience a notable decrease over an extended time period, in contrast to the geographic-economic distance weighting, while the convergence rate of the western region demonstrates an apparent increase. This suggests that the spillover effect of the western region under the spatial correlation weights carries

Table 6 Estimation results of spatial convergence for the regional HQEDI

Variables	Geographic-economic distance weights matrix				Spatial correlation weights matrix			
	East	Central	West	Northeast	East	Central	West	Northeast
LnEQI	-0.1413** (-2.40)	-0.3463*** (-3.80)	-0.2347*** (-3.59)	-0.2170*** (-2.64)	-0.1356** (-1.97)	-3023*** (-3.38)	-0.2695*** (-4.30)	-0.1242*** (-2.88)
EOL	-0.0027 (-0.58)	-0.0030 (-0.056)	-0.0019 (-0.11)	-0.1086** (-2.39)	-0.0026 (-0.06)	-0.0200 (-0.63)	-0.0306* (-1.72)	-0.0306 (-0.79)
ML	0.0010 (0.13)	0.0135 (0.80)	0.0074 (0.55)	0.1914*** (2.69)	0.0021 (0.30)	0.0231* (1.68)	0.0002 (0.02)	0.0436 (0.85)
LWAGE	-0.0055 (-0.22)	-0.0020 (-0.05)	-0.0301 (-0.88)	-0.3606** (-2.24)	-0.0095 (-0.54)	-0.0162 (-0.51)	-0.0495* (-1.78)	-0.0899* (-1.71)
ETL	0.0062*** (3.07)	0.01740*** (2.60)	0.0031*** (2.59)	-0.0454** (-2.17)	0.0045*** (2.64)	0.0146*** (3.17)	0.0071*** (3.58)	-0.0282** (-2.34)
IDL	-0.0009 (-0.43)	0.0184* (1.88)	0.0034 (0.59)	-0.0149 (-0.94)	-0.0009 (-0.32)	0.0162** (2.30)	0.0049 (0.95)	0.0050 (0.55)
PGL	-2.5101** (-2.25)	-1.4036** (-2.45)	-1.1714*** (-2.66)	-0.7961 (-0.15)	-3.3017*** (-2.61)	-1.5037** (-2.09)	-1.4947** (2.34)	-3.8581** (-2.44)
λ	0.6710*** (11.12)	0.1780** (1.97)	0.2599** (2.09)	0.2387*** (3.29)	0.4805*** (7.57)	0.4114*** (4.30)	0.1981*** (2.59)	0.2434** (2.05)
Convergence	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Rate	0.010	0.028	0.018	0.016	0.010	0.024	0.021	0.009
Log L	564.47	341.69	565.06	159.34	552.51	348.72	567.22	156.68
AIC	-1098.94	-653.39	-1100.12	-288.69	-1075.02	-667.44	-1104.44	-283.36
BIC	-1053.78	-615.89	-1053.54	-261.59	-1029.86	-629.94	-1057.86	-256.26

*, **, *** indicate significant at the 10%, 5%, and 1% levels, respectively

more significance under the spatial correlation weights, while the spatial correlation effect present in the eastern region appears relatively weaker.

Regarding the control variables, the influence of regional openness level, marketization level, per capita living standard, and industrial development level on the convergence of the regional HQEDI aligns with the national level in terms of direction, albeit without a significant impact observed for the most part. This can be attributed to the limited variation among regional levels of import and export, fiscal expenditure, employee income, and industrial structure, thus making them ineffective in explaining the spatial coordination of regional high-quality development. The resource transformation level has a negative effect on convergence in the eastern, central, and western regions, suggests that the consistent reduction in energy consumption per unit can positively drive the enhancement of economic quality and industrial efficiency in these regions. The substantial negative impact of local government governance on convergence in the four regions suggests that improving local government governance performance and continuously promoting integrity in government organizations are urgent regional concerns.

6 Conclusions and policy implications

This study dynamically measures the level of high-quality development among China's provinces and examines the spatial convergence of China's HQEDI, thereby unveiling the intricate spatial interaction patterns that characterize regional high-quality development.

Findings illustrate that: the overall level of high-quality development in China experienced a consistent growth from 2002 to 2019, while exhibiting a pronounced regional imbalance trend. The development level declined from the eastern to the western regions, while the growth rate gradually accelerated from the eastern to the western regions, indicating a catching-up effect. The spatial correlation of China's high-quality development level demonstrated a holistic characteristic, indicating that high-quality economic development is a composite of contributions from all regions. The spatial correlation structure revealed unbalanced characteristics. Specifically, Shanghai and Beijing serve as the center, the eastern coastal areas as the sub-center, the central and western regions as subordinate. Further, China's high-quality development exhibited significant spatial agglomeration effects and spatial spillover effects.

The findings of this study offer valuable policy insights for China in advancing the high-quality and coordinated development of regional economic and realizing the long-range objectives of basically achieving socialist modernization. (1) In light of the imbalanced and inadequate regional development in China, the policy measures should be formulated based on various dimensions such as innovation-driven development, coordinated development, market governance, green transformation, open economy, and enhancing people's welfare. Implementation of policies should take into account the heterogeneity of high-quality development across different regions and, accordingly, formulate region-specific strategies for upgrading their development quality, thereby facilitating the comprehensive advancement of

high-quality development across diverse regions. (2) The integration of spatial correlation should be recognized as a crucial variable in the strategic formulation of high-quality development policies. Given the variations in core-subordinate relationships across different provinces with regards to spatial correlation, a balanced approach of central coordination and local guidance is essential. Emphasizing precise management within regions and harmonized progress among regions should be the fundamental guiding principles. This includes bolstering inter-provincial and interregional collaboration, actively engaging local stakeholders to stimulate their initiative and enthusiasm, and progressively advancing the optimization of resource allocation and regional integration processes. (3) The formulation and implementation of government policies should effectively tackle the intricate and divergent nature of high-quality development in different regions, fully consider regional demands and escalate the regulation and control of the government's actions. The execution of pertinent policies should aim to boost the diffusion and impetus in the eastern regions and reinforce their spatial externalities, also establish a linkage mechanism among the eastern, central, and western regions to amplify the interaction between the development ideologies and policy formulation of both parties. This will expedite the convergence of the spatial expansion rate of economic development and catalyze the synchronized and high-quality development of regions.

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Declarations

Conflict of interest The authors declared that they have no conflicts of interest in this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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